mmDar Dense Point Clouds via Physics-Informed Set Prediction

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PROBLEM & APPROACH

mmWave radar (77 GHz FMCW) offers all-weather, low-cost 3D sensing but angular resolution is limited: 8 virtual antennas yield ~14° Rayleigh resolution.

The original RadarHD pipeline discards critical information:

- **Phase discarded** — magnitude-only PNGs lose angular encoding
- **Dynamic range reduced** — 60+ dB → 8-bit (48 dB)
- **Weak reflections lost** — 5% magnitude threshold

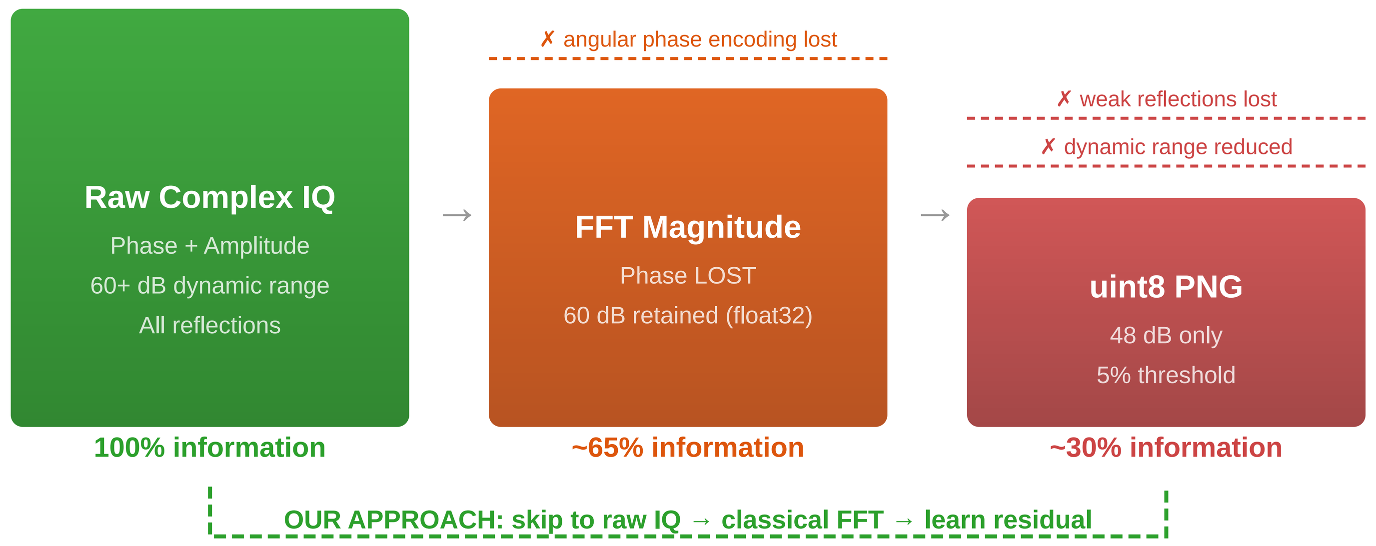
Core Change

Move from *preprocessed magnitude PNGs* to **raw complex IQ data**, making the model **physics-informed**: classical FFT beamforming as input features, network learns only the residual improvement and enable **real-time processing**.

Physics as Input

The FFT beamformer provides the physics baseline. The network learns what physics *cannot* resolve — improving angular localization beyond the Rayleigh limit for sparse targets.

INFORMATION RETAINED AT EACH PROCESSING STAGE



EVALUATION METHODOLOGY

All metrics computed on BEV point clouds (polar → Cartesian). **Checkpoints selected on validation set only**.

Chamfer Distance (Mean Error)

$$CD = \tfrac{1}{2}\left[\, \Sigma \min\|p-q\| + \Sigma \min\|q-p\| \,\right]$$

Average bidirectional nearest-neighbor distance. Measures **coverage**.

Modified Hausdorff (Worst-Case)

$$MH = \max(\, \mathrm{median}(NN_{A\to B}),\ \mathrm{median}(NN_{B\to A}) \,)$$

Median of nearest-neighbor distances, max direction. Penalizes **outliers and gaps**.

IoU & F1 (Polar Grid)

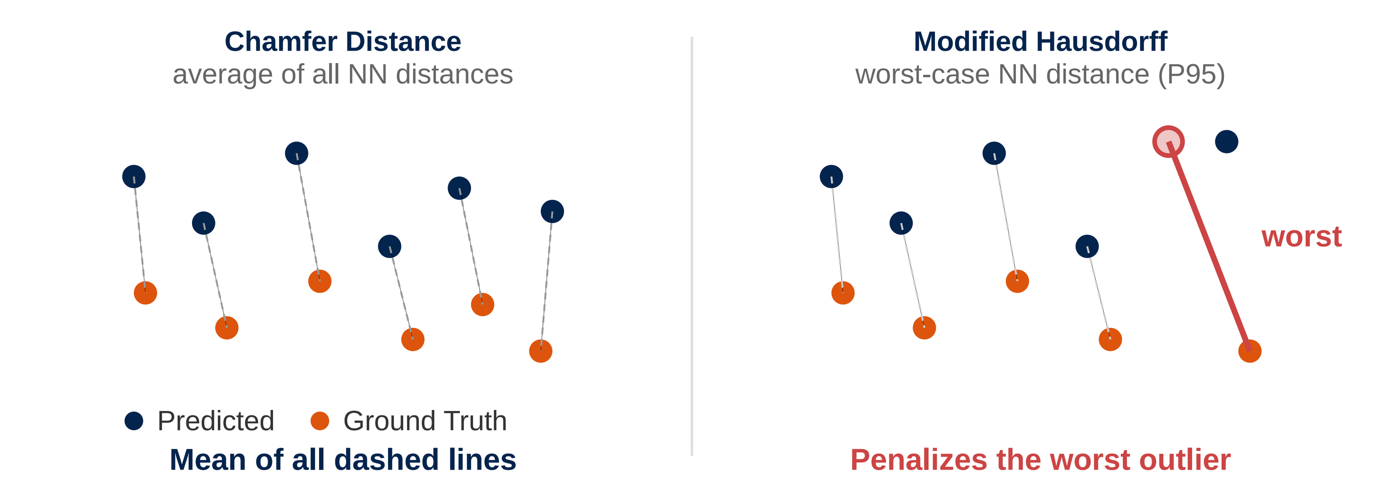
Binary classification on 256×512 polar grid. IoU measures overlap of occupied cells; F1 balances precision and recall of detections.

Precision / Recall @ 0.1m

Precision: fraction of predicted points within 0.1m of any GT point. **Recall:** fraction of GT points with a prediction within 0.1m. Threshold-based point-level accuracy.

Why Multiple Metrics?

Chamfer measures *mean* quality. mod-H measures *worst-case*. IoU/F1 measure *detection*. A model that sprays points scores well on Chamfer but poorly on mod-H. The Gaussian model improves **all metrics simultaneously**.



MODEL ARCHITECTURE ITERATIONS

We systematically explored architectures, moving from preprocessed images to raw radar data with physics-informed design. Each iteration informed the next.

Baseline: Asymmetric U-Net on Magnitude PNGs (RadarHD)

Raw IQ (complex128) → FFT + |·| (magnitude) → uint8 PNG (41 frames) → U-Net (17.5M params) → BEV Points

Final: Physics-Informed Gaussian Set Decoder

Raw IQ (complex64) → Classical FFT (fixed weights) → 2D Encoder (ResBlocks) → DETR Decoder (96 queries) → Gaussian Head (μ, σ per point) → BEV Points

All Iterations & Results

Model / Iteration	Input	Params	Chamfer ↓	Mod-H ↓	Key Finding
Published RadarHD	PNGs	17.5M	0.360	0.240	Original paper result
Baseline (U-Net)	PNGs ×41	17.5M	0.406	0.296	Honest val-selected eval
ConvLSTM Temporal	PNGs seq.	27.5M	0.603	—	Temporal fusion too late; 2× worse
Raw IQ Mag+Phase	IQ ×1	~2M	0.309	0.423	Phase helps Chamfer; single-frame limit
Temporal Cross-Attn	FFT ×8	2.2M	0.295	0.429	Matched baseline Chamfer; mod-H gap
Physics-Informed Loss	FFT	—	14.5	—	Diverged — competing gradients
Gaussian (8 frames)	IQ ×8	1.6M	0.421	0.345	Hungarian NLL > Chamfer for mod-H
Gaussian (41 frames)	IQ ×41	4.6M	0.356	0.278	Temporal context adds 19% mod-H gain
Physics-First 2D Gaussian	FFT 2D ×41	3.1M	0.318	0.230	Best: FFT as input + DETR + Gaussian
LISTA + U-Net Occupancy	FFT features	~3M	1.281	1.844	Beamformer is information bottleneck

Final Architecture: Physics-First 2D Gaussian Set Decoder

ENCODER PATH

Raw Complex IQ Input
(8 antennas, 512 range bins) complex64 × 41 frames

↓

Classical FFT Beamformer (FIXED)
64 azimuth bins — no trainable weights — physics as input
→ (82, 64, 512)

↓

Deep 2D Encoder

- **ResBlock2d + GroupNorm** — (64, 512) → 128 channels
- stride-2 ↓
- **DownBlock (32, 256) → 128ch** — Conv2d + GroupNorm + GELU
- stride-2 ↓
- **DownBlock (16, 128) → 128ch**
- stride-2 ↓
- **DownBlock (8, 64) → 128ch** — Spatial features: 8 × 64 × 128

DECODER PATH

Gaussian Prediction Head (per query)

μr, μφ — position	existence logit
σr, σperp — uncertainty	σperp = f(range)

↑

DETR-Style Set Decoder

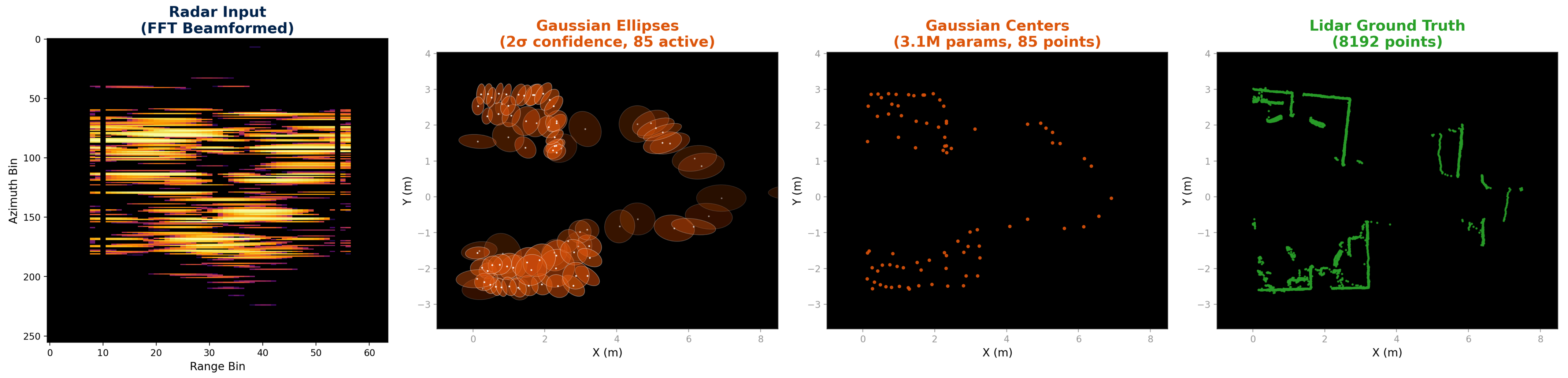
- 96 Learnable Object Queries (randomly initialized)
- **Multi-Head Cross-Attention Layer 1** — Q: queries (96) K,V: spatial tokens (512)
- **Multi-Head Cross-Attention Layer 2** — + Feed-Forward Network + LayerNorm
- **Multi-Head Cross-Attention Layer 3** — Output: 96 refined query embeddings

Loss: Hungarian-matched heteroscedastic NLL

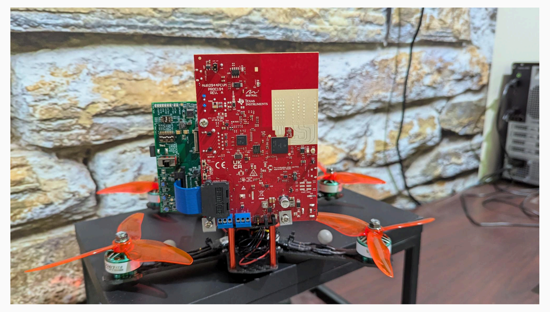
Flatten + 2D Positional Encoding
8×64 = 512 spatial tokens, each 128-dim — 2D sinusoidal PE preserves spatial structure

3.1M total parameters (5.6× smaller than baseline)

Visual Comparison



DATASET

Paired **TI AWR1843** 77 GHz FMCW radar + **Velodyne VLP-16** lidar, synchronized per-frame (RadarHD dataset).

42k PAIRED FRAMES	**44** TRAJECTORIES
8 VIRTUAL ANTENNAS	**512** RANGE BINS

Split: 17 train / 8 val / 19 test trajectories
Test set: 18,575 frames — checkpoint selected on validation set only

Input format: Raw complex IQ (8 antennas × 512 range bins, complex64) after range FFT and TDM-MIMO Doppler compensation. 41-frame temporal stacking.

Ground truth: Lidar **BEV** (Bird's-Eye View) point clouds projected to polar coordinates for direct comparison.

DATA PROCESSING PIPELINE

TI AWR1843 (77 GHz FMCW, 4 RX × 2 TX) → Range FFT (512 bins, complex64) → TDM-MIMO Compensation (→ 8 virtual ant.) → Model Input ((8, 512) complex64 × 41 frames)

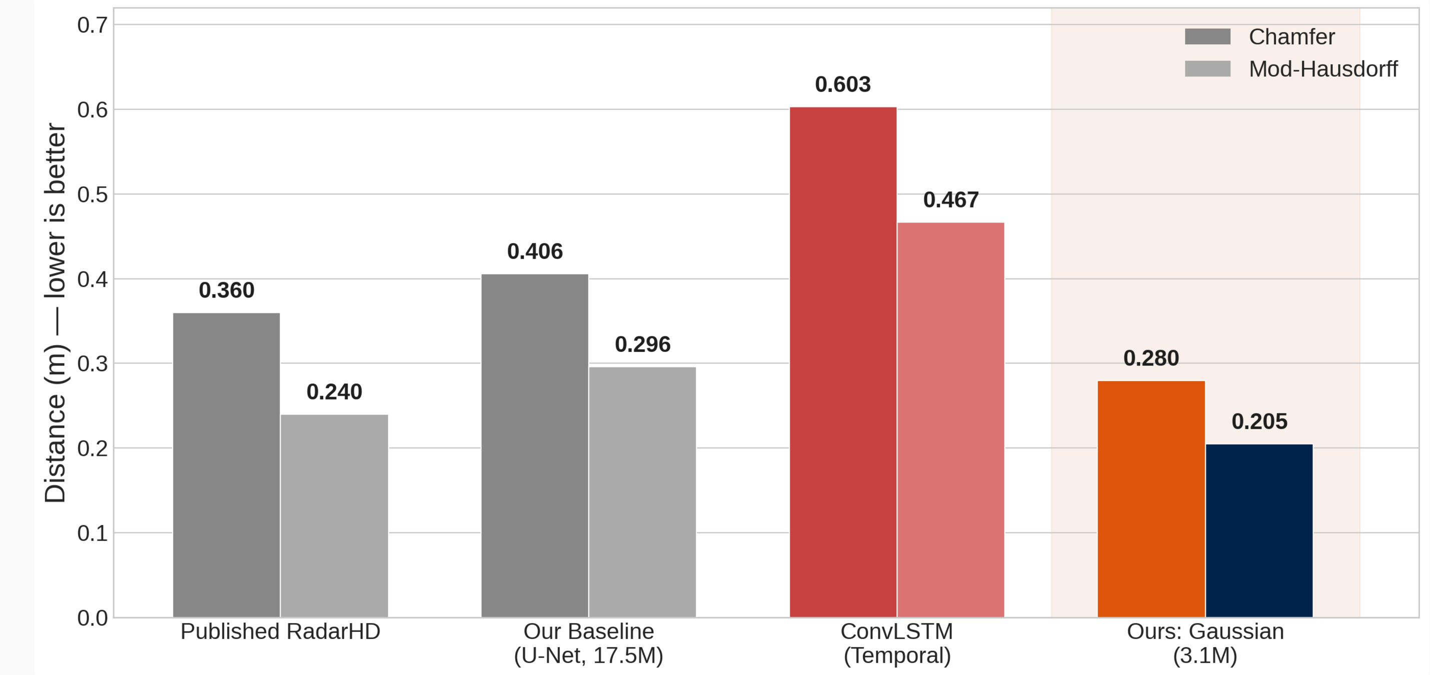
SYNCHRONIZED PER-FRAME

Velodyne VLP-16 (16-beam lidar, 100m range) → BEV Projection (3D → 2D top-down) → Polar Transform (256 az × 512 range) → Ground Truth (BEV point cloud in polar coords)

RESULTS & CONCLUSIONS

Model	Chamfer ↓	Mod-H ↓	Params
Published RadarHD	0.360	0.240	17.5M
Baseline (U-Net)	0.406	0.296	17.5M
ConvLSTM Temporal	0.603	—	27.5M
mmDar (Gaussian)	0.318	0.230	3.1M

22% Lower Chamfer & mod-H vs. honest baseline	**5.6×** Fewer parameters (3.1M vs 17.5M)	**Raw IQ** vs. preprocessed PNGs preserves phase info



Conclusions: Operating on raw complex IQ instead of preprocessed PNGs, our 3.1M-parameter model achieves 22% lower error than the 17.5M-parameter baseline at **<3ms per frame** — well under 50ms real-time. mmWave radar with physics-informed architectures can produce dense point clouds approaching lidar quality at a fraction of the sensor cost.

Selected Sources: [1] Rebut et al., "RadarHD," ICRA 2023 [2] Zhu et al., "mmPoint," BMVC 2023 [3] Carion et al., "DETR," ECCV 2020 [4] TI AWR1843 Technical Reference